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Introduction:

Software quality processes ensure that software works correctly, performs efficiently, protects data, and remains secure.

In software industry, many of these processes are automated using testing tools, CI/CD pipelines, and AI-powered agents to detect problems quickly and reduce manual effort.

The four most critical software quality processes are:

- Functional Testing
- Performance Testing
- Data Quality Testing
- Security Testing

Functional Testing

Definition: Functional testing checks whether software features work according to the specified

Example: In a university portal, functional testing checks whether

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student can log in, register for courses, and view their results correctly.

Testing activity	Automation tools/agents
Unit testing	JUnit, TestNG, pytest
Web UI testing	Selenium, Playwright, Cypress
API testing	Postman, REST Assured
AI-assisted test creation	GitHub Copilot & other AI coding assistants

Performance Testing

Definition: Performance testing checks how fast, stable, and responsive software remains under different workloads

Example: A university portal is tested to determine whether it can handle 10,000 students logging in simultaneously without becoming slow or crashing.

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Testing activity	Automation tools/agents
Load and stress testing	Apache, JMeter, Grafana k6
API performance testing	k6, JMeter
Browser performance testing	Lighthouse, k6 Browser
Performance monitoring	Grafana, New Relic

These tools support automated load tests, repeated performance checks, and monitoring.

What is automated?

Simulating thousands of virtual users.
Measuring response time and throughput.
Finding bottlenecks under heavy traffic.
Checking whether performance meets defined limits.

Data Quality Testing

Definition: Data quality testing checks whether data is accurate, complete, consistent, valid and free from unexpected duplicates.

Example: In a University database, testing verifies that every student has a valid student ID, required fields are not empty and student records

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are not duplicated.

Testing activity	Automation tools/agents
Data validation	Great Expectations
Database and data pipeline testing	dbt tests
Data quality monitoring	Soda
Database integrity checks	SQL scripts and constraints

These tools automate data validation and can identify problems during data processing or pipeline execution.

What is automated?

- Checking missing or duplicate values.
- Validating data types and formats.
- Verifying database relationships and constraints.
- Detecting unexpected changes in data.
- Checking data pipelines after new data arrives.

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Security Testing

Definition: Security testing identifies vulnerabilities and verifies that software protects its users, systems and data against unauthorized access and attacks.

Example: A university portal is tested to ensure that a student cannot access another student's marks by changing a URL or request.

Testing activity	Automation tools / agents
Source code security analysis (SAST)	SonarQube, CodeQL
Web application security testing (DAST)	OWASP ZAP, Burp Suite
Dependency vulnerability scanning (SCA)	Dependabot, Snyk
Container and configuration scanning	Trivy

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Security tools can be integrated into CI/CD pipelines to detect common vulnerabilities before deployment.

What is automated?

- Detecting insecure coding patterns.
- Finding known vulnerabilities in dependencies.
- Scanning web applications for common security weaknesses.
- Identifying insecure configurations.
- Alerting developers when security checks fail.